

Adaptation:

Perhaps the most interesting feature of Atick & Redlich's theory is that it predicts an important biological phenomenon, adaptation:

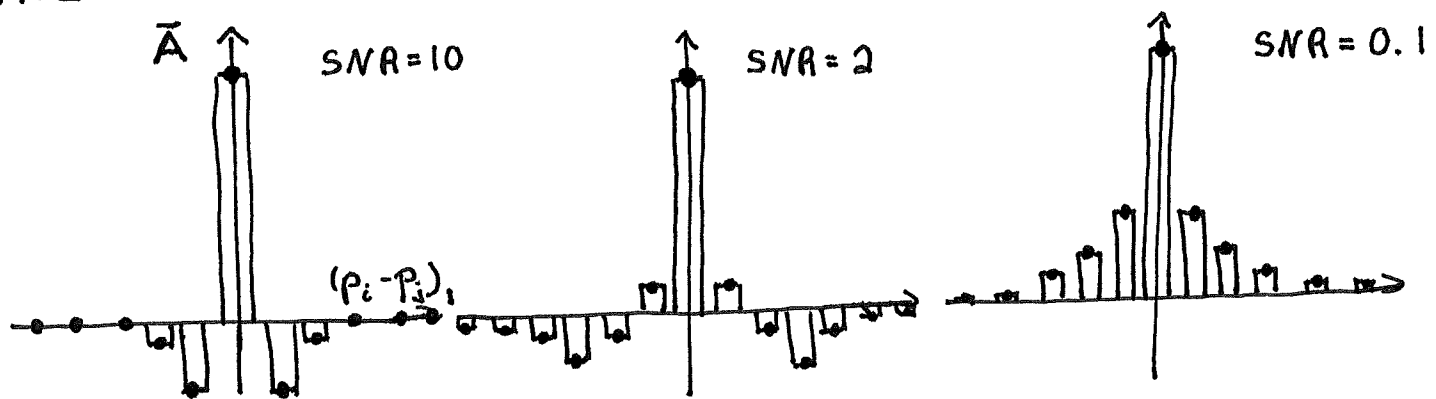
Recall the input statistics:

$$(C_x)_{ij} = \underbrace{S^2 e^{-\|p_i - p_j\|/D}}_{\text{Signal}} + \underbrace{\sigma_x^2 \delta_{ij}}_{\text{Noise}}$$

Signal to noise ratio (SNR):

$$SNR = \frac{S}{\sigma_x}$$

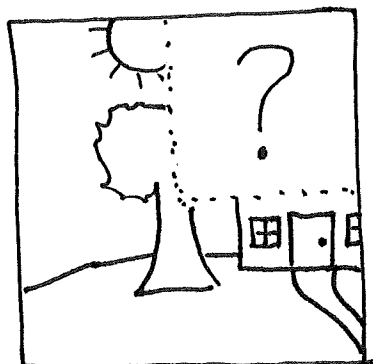
The SNR set the relative contributions of the center and surround



At high SNR, decorrelation was achieved by a strong surround that subtracts out correlated signals from the center. At low SNR, noise reduction was achieved by averaging the signal over space. Redundancy remains because of the constraint that $I(s; y) = I^*$. Intermediate SNR interpolates between these cases.

Natural scene statistics:

Images of the natural world are highly structured and correlated. What's missing below?



Precisely quantifying these statistics can improve our understanding of visual processing and sharpen efficient coding theories.

Correlation function:

If you translate a natural image, you get another natural image. This "translational invariance" means that the correlation of a signal between two points depends only on the ~~displacement~~ ^{displacement} between them.

$$C_s(p_1, p_2) = C_s(p_1 - p_2) = C_s(\delta p), \quad p_2 = p_1 + \delta p$$

Fourier ~~to~~ transform:

The Fourier transform rewrites a function of space as a function of "spatial frequencies."

$$f(x) = \int_{\mathbb{R}^n} d\xi \, e^{i2\pi \xi \cdot x} \hat{f}(\xi)$$

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} dx \, e^{-i2\pi \xi \cdot x} f(x)$$

$\mathbb{R}^n$ will usually leave out of notation

The relationship to spatial frequencies follows from the fact that

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Thus

$$e^{i 2\pi \xi \cdot x} = \underbrace{\cos(2\pi \xi \cdot x) + i \sin(2\pi \xi \cdot x)}_{\text{Sinusoidal functions with } \xi \text{ cycles per } 360^\circ (2\pi \text{ rad})}$$

vector in $\mathbb{R}^n$, ξ_x, ξ_y , etc.

Don't be scared by the complex numbers. Everything could be done with sinusoids, but ^{that's} more cumbersome notationally and ultimately harder.

A useful feature of the Fourier transform is that it transforms convolutions into products.

$$\begin{aligned} \widehat{(f * g)}(\xi) &= \int dx e^{-i 2\pi \xi \cdot x} (f * g)(x) \\ &= \int dx e^{-i 2\pi \xi \cdot x} \int dx' f(x') g(x - x') \\ &= \int dx' f(x') \underbrace{\int dx e^{-i 2\pi \xi \cdot x} g(x - x')}_{\int dx'' e^{-i 2\pi \xi \cdot (x' + x'')} g(x'')} \\ &= \int dx' e^{-i 2\pi \xi \cdot x'} f(x') \int dx'' e^{-i 2\pi \xi \cdot x''} g(x'') \\ &= \hat{f}(\xi) \hat{g}(\xi) \end{aligned}$$

For receptive field models:

$$r(x) = (K * s)(x) \Rightarrow \hat{r}(\xi) = \hat{K}(\xi) \hat{s}(\xi)$$

It's therefore convenient to specify receptive fields in the frequency domain.

Power spectrum:

Let's Fourier transform the correlation function.

$$\hat{C}_s(\xi) = \int d(\delta p) e^{-i 2\pi \xi \cdot \delta p} C_s(\delta p)$$

Analysis of 2D natural images suggests that this has a very simple form

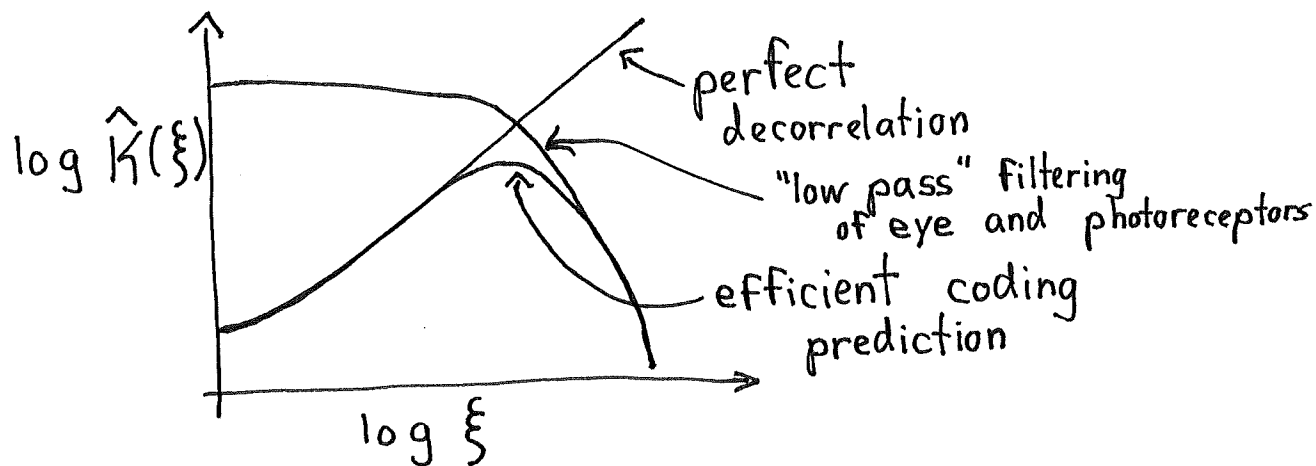
$$\hat{C}_s(\xi) \sim \frac{1}{|\xi|^2}$$

Back to decorrelation:

For a perfectly decorrelated code,

$$C_{xy}(\delta p) = B \delta(\delta p) \Rightarrow \hat{C}_y(\xi) = \int d(\delta p) e^{-i 2\pi \xi \cdot \delta p} B \delta(\delta p) = B$$

The receptive field that decorrelates natural images is thus the one that produces a uniform power spectrum at the output level. In a follow up 1992 paper, Atick & Redlich computed the receptive field in this form, with $\hat{C}_s \sim 1/|\xi|^2$



As with their earlier theory $\hat{K}(\xi)$ adapts to SNR (here light level). The predicted form matched data well across light levels.

Adaptation in LVP models:

To estimate adaptation in LVP models, one can use white noise to fit them at different light levels (thus, noise levels). One finds these adaptation effects.

More generally, one can examine how the LVP model varies with the stimulus statistics used for fitting.